

THE CAUSAL CLOSED LOOP OF PHYSICAL AI

From Digital Policy Deliberation across the Causal Boundary to Physical Action and Endogenous Observation Shift

UNTRUSTED COMPUTATIONAL REALM · APPLICATION PROCESSOR (HOST LINUX MPU / ACCELERATOR)

IDEMPOTENT · REVERSIBLE · SOFTWARE CHECKPOINTS

1. Sensory Ingestion

Multi-Camera / IMU / Encoders
Observation Vector: $\mathbf{o}_t$ in $\mathbb{R}^d$
Spatial Tokenization & Calibration

tokens

2. Neural Policy Deliberation

VLA / Diffusion Policy / ACT
Stochastic Candidate: $\hat{\mathbf{a}}_t \sim \pi_{\theta}(\mathbf{o}_t)$
Unprivileged Candidate Proposal

proposal

3. Deterministic Safety Referee

Control Barrier Function (CBF-QP)
Forward Invariance: $\dot{h} + \gamma h \geq 0$
Permission / Fallback Safe Stop

THE CAUSAL BOUNDARY: MEMORY-MAPPED ACTUATOR REGISTER WRITE

$\mathbf{u}_t$ (committed)

PHYSICAL REALITY · DYNAMICS & THERMODYNAMICS (ACTUATOR INVERTERS, MASS, ENVIRONMENT)

NON-REVERSIBLE · THERMODYNAMIC WORK · JOULE HEAT

4. Power Inverter & Windings

Gate Driver PWM Switching
Joule Heat: $Q = I^2 R \Delta t$
Magnetic Flux & Lorentz Force

torque τ

5. Mechanical Dynamics

Torque $\tau = M(q)\ddot{q} + C(q, \dot{q})$
Kinetic Momentum: $\mathbf{p} = m\mathbf{v}$
Reflected Inertia: $N^2 J_{\text{rotor}}$

work $F dx$

6. Physical Environment

Environment State: $W_t \rightarrow W_{t+1}$
Delay as Distance: $\Delta x = \int v dt$
Unrecoverable State Transitions

ENDOGENOUS SENSORY FEEDBACK: $\mathbf{o}_{t+1} \sim P(\mathbf{O} | \mathbf{s}_{t+1}, W_{t+1})$ [Actions reshape future observations]